Unit: 6 Unit 8: Quadratics
Topic: Vertex Form and Intercept Form

Name: ______________
Date: __________

Quiz (Habanero)

Q1. A quadratic function models the CO_2 levels in a forest ecosystem. The function is $f(x) = (x-3)^2 + 2$, where x is the number of years since 2020. What is the vertex form of this function?
(A) $(h, k) = (3, 2)$
(B) $(h, k) = (2, 3)$
(C) $(h, k) = (1, 4)$
(D) $(h, k) = (4, 1)$
(E) $(h, k) = (5, 0)$

Q2. The temperature T in a region is modeled by the quadratic function $T(x) = x^2 - 4x + 4$, where x is the number of hours since sunrise. Write this function in intercept form.
(A) $T(x) = (x-2)(x+2)$
(B) $T(x) = (x-2)(x-2)$
(C) $T(x) = (x+2)(x+2)$
(D) $T(x) = (x-1)(x-3)$
(E) $T(x) = (x+1)(x-3)$

Q3. A quadratic equation $x^2 + 5x + 6 = 0$ models the growth of a plant species in a controlled environment. What are the x-intercepts of this equation?
(A) $(-2, 0)$ and $(3, 0)$
(B) $(-3, 0)$ and $(2, 0)$
(C) $(-1, 0)$ and $(6, 0)$
(D) $(-6, 0)$ and $(1, 0)$
(E) $(-4, 0)$ and $(5, 0)$

Q4. A researcher is studying the pH levels in a lake ecosystem. The pH level P is modeled by the quadratic function $P(x) = (x-2)^2 + 1$, where x is the amount of rainfall in inches. What is the minimum pH level?
(A) $P(2) = 1$
(B) $P(1) = 2$
(C) $P(3) = 3$
(D) $P(4) = 4$
(E) $P(5) = 5$

Q5. The quadratic function $f(x) = x^2 - 3x - 4$ models the population growth of a bird species in a forest. Write this function in vertex form.
(A) $f(x) = (x-1.5)^2 - 6.25$
(B) $f(x) = (x-2)^2 - 5$
(C) $f(x) = (x+1.5)^2 - 6.25$
(D) $f(x) = (x+2)^2 - 5$
(E) $f(x) = (x-3)^2 - 4$

Q6. A quadratic equation $x^2 - 2x - 3 = 0$ models the wind speed in a region. What are the x-intercepts of this equation?
(A) $(-1, 0)$ and $(3, 0)$
(B) $(-3, 0)$ and $(1, 0)$
(C) $(-2, 0)$ and $(2, 0)$
(D) $(-4, 0)$ and $(2, 0)$
(E) $(-5, 0)$ and $(1, 0)$

Q7. A quadratic function $f(x) = (x+1)^2 - 2$ models the water level in a lake. What is the vertex of this function?
(A) $(-1, -2)$
(B) $(-2, -1)$
(C) $(-1, 2)$
(D) $(1, -2)$
(E) $(2, 1)$

Q8. The quadratic function $f(x) = x^2 + 2x - 3$ models the temperature changes in a region. Write this function in intercept form.
(A) $f(x) = (x+3)(x-1)$
(B) $f(x) = (x-3)(x+1)$
(C) $f(x) = (x+1)(x-3)$
(D) $f(x) = (x-2)(x+1)$
(E) $f(x) = (x+2)(x-1)$

Open Ended Questions (FRQ)

Q9. Suppose a quadratic function $f(x) = ax^2 + bx + c$ models the NO_x emissions in a city, where x is the number of vehicles. If the vertex of the function is $(-2, 100)$, write the function in vertex form and find the value of a if $f(0) = 150$.

Q10. A researcher is studying the effect of CO_2 levels on plant growth. The growth G is modeled by the quadratic function $G(x) = (x-1)^2 + 2$, where x is the amount of CO_2 in parts per million. Find the x-intercepts and the vertex of the function, and interpret the results in the context of the problem.

Chef's Tips

- Tip 1: Ensure students understand the concept of vertex and intercept forms of quadratic functions.
- Tip 2: Emphasize the importance of interpreting parameters in each form.
- Tip 3: Provide feedback on the uniqueness and accuracy of student responses.